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U_CAN

Ukraine towards Carbon Neutrality

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Ukraine towards Carbon Neutrality

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1. EXPLANATION OF THE WORK CARRIED OUT AND OVERVIEW OF THE PROGRESS

Ukraine towards Carbon Neutrality (U_CAN) is a Coordination and Support Action funded under the EU Framework Programme for Research and Innovation, Horizon Europe, and contributes to the EU Mission Climate-Neutral and Smart Cities. U_CAN supports selected Ukrainian cities in aligning their strategies and standards with EU climate policies. This is achieved through targeted support to Ukrainian city administrations in their efforts to collaborate with European counterparts, including EU Mission Cities.

This section provides an overview of the work carried out and progress achieved during the first reporting period (months 1–18) of the U_CAN project. Progress is reported against the project objectives as defined in the Description of Action (DoA) and focuses on activities implemented to date. The section introduces the main areas of work, including twinning activities, pilot preparation, capacity building, impact, and dissemination, which are described in more detail in the subsequent subsections.

1.1 Objectives

Objective 1: to support Ukrainian cities that are willing to transition to climate neutrality by developing context specific climate strategies. The envisioned outcome is a core group / network of pioneering Ukrainian stakeholders who require support in building capacities to address climate neutrality.

Progress:

- Six Ukrainian cities (Lviv, Kyiv, Zhytomyr, Khmelnytskyi, Ivano-Frankivsk, Vinnytsia) were engaged as a structured pilot group for climate-neutral transition activities. Lviv, Zhytomyr, and Vinnytsia participate through direct municipal representatives, while Kyiv, Khmelnytskyi, and Ivano-Frankivsk are represented through university partners and NGOs respectively. In all cases, designated municipal contact persons were identified, involved in all key workshops, and engaged in bilateral discussions with project partners.
- City-specific needs, priorities, and institutional capacities were collected through structured exchanges involving representatives from each pilot city, including Knowledge and Experience Exchange Sessions (KEES) (2 synergy workshops and over 20 bilateral meetings), 2 extensive questionnaires, 3 thematic workshops led by Mission Cities, 5 structured interviews, and 2 dedicated workshops. These activities were implemented by multiple project partners across different research areas, as detailed in the respective Work Package descriptions.

Objective 2: to identify key sectors in Ukraine for climate transition. This includes identifying critical sectors that require transitioning including industrial decarbonisation, energy transition, nature-based solutions, etc. Outcomes envisioned here are recommendations, standards and best practices co-developed with EU experts.

Progress:

- Key climate transition sectors were identified at national and municipal levels through a combination of policy analysis, scientific literature review, and structured inputs from the six Ukrainian pilot cities. This work resulted in the completion of *Deliverable D3.1 – Sectors for Climate Actions in Ukraine*, which provides an integrated assessment of priority sectors, policy relevance, local capacities, and barriers.
- Based on this sectoral mapping, research tasks under WP5 have been initiated and advanced, analysing standardisation, modernisation, and technological needs across the identified sectors. This includes the completion of *Deliverable D5.1 – Overview of the standardisation landscape* and ongoing analytical work on building renovation, energy systems, industrial

decarbonisation, and local climate governance constraints, providing a concrete basis for translating sectoral priorities into technical and procedural pathways.

Objective 3: to facilitate CNSC twinning activities, fostering the exchange of best practices in climate action. This facilitation includes connecting CNSC Mission cities as pioneering examples and mentors for Ukrainian cities to achieve climate neutrality. This leadership will leverage the expertise and success stories of established CNSC cities to inspire and support their Ukrainian counterparts. The envisioned outcomes are long-term partnerships between CNSC Mission cities and partner Ukrainian cities.

Progress:

- Partnerships were formally established with six Mission Cities (Bratislava, Dresden, Košice, Leipzig, Thessaloniki, and Wrocław), to support our Ukrainian pilot cities in their transition to climate neutrality.
- A structured twinning framework between Ukrainian pilot cities and EU CNSC Mission Cities was established and operationalised through the design and implementation of the Knowledge and Experience Exchange Sessions (KEES).
- Twinning activities were implemented through multiple exchange formats, including 2 CNSC Synergy Workshops (1 online and 1 hybrid at Wrocław), over 20 bilateral meetings between Ukrainian and Mission City representatives, and 3 thematic workshops addressing priority topics such as sectoral pathways, solar and low-carbon solutions, and public engagement. These activities enabled targeted knowledge transfer, needs clarification, and the assessment of compatibility for longer-term city-to-city cooperation.

Objective 4: to facilitate local Pilot initiatives as indication of on-ground efforts demonstrating feasibility and benefits of climate-neutral measures. This will also propagate tangible efforts in terms of replication and knowledge exchange between Ukrainian partner cities and other cities in Ukraine. Outcomes envisioned here are localised pilot implementation plans.

Progress:

- Task 4.1 progressed through the development and application of a common framework for climate-neutrality indicators in pilot cities. A preliminary set of indicators was defined using a combined methodological approach, drawing on international sustainability assessment frameworks and adapted to the Ukrainian context. Indicators were reviewed and prioritised by representatives of all six pilot cities based on relevance and data availability, supported by workshops, questionnaires, and bilateral meetings. By the end of the reporting period, Milestone MS5 (Indicators for climate neutrality in the context of pilot activities) was achieved, and the preparation of *Deliverable D4.1* was underway.
- Across Tasks 4.2–4.7, pilot implementation activities were initiated in all six pilot cities, focusing on city-specific priority areas and the preparation of local pilot concepts, in coordination with municipal stakeholders and European partners. These activities prompted the gradual development of the pilot concepts, institutional arrangements, and stakeholder engagement processes that form the basis for localised pilot implementation plans.

Objective 5: to facilitate awareness-building among a wider group of Ukrainian cities. This will ensure the dissemination and exploitation of project results for long-term impacts on Ukraine's alignment with the EGD. Outcome envisioned here is exploitation of the project activities to pan-Ukraine CNSC initiatives.

Progress:

- Project results and activities were actively disseminated to a wider group of Ukrainian and European stakeholders through the establishment and operation of the U_CAN project platform and associated communication channels. During the reporting period, the bilingual project website was launched and regularly updated, supported by interactive maps, news

items, newsletters, and social media channels (LinkedIn, Facebook, YouTube), enabling access to project information beyond the project partners and the six pilot cities.

- Communication and dissemination materials (brochures, postcards, digital banners, videos) were developed and used to present U_CAN objectives, pilot city activities, and emerging results, strengthening visibility and engagement with Ukrainian cities beyond the core pilot group.
- Engagement with the EU Climate-Neutral and Smart Cities Mission ecosystem was initiated and advanced. This involved the formalisation of an MoU with NetZeroCities, the creation of a public group within the NetZeroCities Platform to share U_CAN activities and outputs, and active participation in Cities Mission–related events, including hosting a Ukraine Cluster stand at the Cities Mission Conference. These actions increased the visibility of U_CAN and supported outreach to Mission Cities beyond the six partner Mission Cities involved in the project.

#@WRK-PLA-WP@#

1.2 Explanation of the work carried out per WP

1.2.1. Work Package 1: Project Management (M1-M48, TUD)

WP1 Objectives

The aim of this WP is to ensure effective management of the project with emphasis on meeting the European Commission requirements. The objectives are:

- Management and monitoring of project progress of WPs against deliverables.
- Ensuring accurate, timely reporting on activities and communication with the EC.
- Ensuring Compliance and the fulfilment of ethical necessities and
- Exchange with Network Partners.

Progress in WP1 during the first Reporting Period of the project

WP1 focused on establishing the organisational, administrative, and coordination frameworks necessary for the effective implementation of U_CAN. Key management structures, procedures, and processes were set up to support consortium collaboration, ensure compliance with administrative and reporting requirements, and facilitate engagement with external stakeholders.

Table 1: Overview of deliverables (D) & milestones (MS) of WP1

#	Title	Due date	Actual delivery	Delay justification
D1.1	Project Management Handbook	31/07/2024	09/09/2024 (20/01/2025)	Request by CINEA to include collaboration with SUN4Ukraine and integration of CINEA feedback following 1st submission
D1.2	Data Management Plan	31/10/2024	01/11/2024 (20/01/2025)	Integration of CINEA feedback following 1st submission
D1.3	1st update of Data Management Plan	31/10/2025	10/11/2025	Long negotiations on joint control- lership agreement
MS1	Working Network Partners Board established	31/07/2024	12/08/2024	Late confirmation of some network partners and small delay with completion of interactive map of network partners

Task 1.1: Management of Consortium and Work Packages

M1-M48; TUD (WP-Leads, all partners)

Achievements:

Q1. With a consortium of 21 partners, the first months of the project were used to establish meeting protocols and a solid management structure, defining the roles of the Coordinator, the Administrative Manager, WP-Leaders, Task-Leaders and partner representatives. Following several online consortium meetings prior to the project start, the face-to-face Kick-off meeting took place on 23-24 May 2024 in Dresden, Germany. This event brought together all consortium partners to outline the project objectives and initial work plan, align expectations, and lay the groundwork for effective collaboration.

Q2-Q4. To ensure continuous communication and coordination, monthly online Work Package meetings and bi-monthly online Consortium Meetings were facilitated by TUD as the main coordination frameworks. These meetings supported the regular exchange of knowledge and results, identification of synergies, and discussion of potential challenges and upcoming activities.



Figure 1: U_CAN Consortium Meeting at (a) Dresden, Germany (left), (b) Wrocław, Poland

Q5. A second in-person Consortium Meeting was held on 21–23 May 2025 in Wrocław, Poland, serving as a Year 1 review and Year 2 planning meeting. The programme was structured as follows: the first day focused on general Work Package progress updates, the second day was dedicated to eight parallel workshops, and the third day was devoted to the Synergy Workshop between Ukrainian Cities and EU Mission Cities.

Q6. Report preparation for the first reporting period was coordinated at consortium level. An internal workshop titled “Get Ready for EU Reporting” was organised by TUD to support partners in understanding reporting requirements and preparing technical and administrative inputs.

Deviations and Impact on Other Tasks or Work Packages:

None.

Task 1.2: Administration and Financial Management

M1-M48; TUD-EPC (all partners)

Achievements:

Q1. The Consortium Agreement, a vital document defining the roles, responsibilities, and rights of all partners, was successfully signed. This agreement provides a legal framework for the collaboration, outlining governance structures, intellectual property rights, and conflict resolution mechanisms. Furthermore, the first part of the pre-financing was distributed by the Coordinator TUD to all partners.

Q2. A comprehensive Project Management Handbook (D1.1) was drafted by TUD, detailing the project's management structure, procedures, and guidelines. This document serves as a reference for all partners, ensuring consistency and clarity in project operations.

Q3-Q4. To facilitate effective collaboration, a suite of tools and platforms has been implemented by TUD. These include the Project Management Tables (complementing the Project Management Handbook), a shared document repository, and mailing lists. These tools have streamlined workflows, enhanced information sharing, and supported remote collaboration, especially amidst the current challenging scenario in Ukraine. In addition, an internal financial report was prepared by all partners, allowing the Coordinator TUD to get an overview of current spending.

Q5-Q6. Financial documentation from all partners for the first reporting period was coordinated, reviewed, and consolidated. Emerging issues were identified and resolved in coordination with

partners to ensure timely and accurate financial reporting. During an internal workshop on 07.10.2025 the Coordinator TUD explained the details of financial reporting and calculation of eligible costs to all consortium partners.

Deviations and Impact on Other Tasks or Work Packages:

Deliverable D1.1 was resubmitted following the incorporation of feedback from CINEA. The required revisions were minor and, as a result, the adjusted timeline did not have any negative impact on subsequent tasks or work packages.

Task 1.3: Data Management, Ethics and Gender Assurance

M1-M48; TUD (ULA, all partners)

Achievements:

Q1-Q2. The *Deliverable D1.2. Data Management Plan* was developed by TUD, outlining the strategies for data collection, storage, sharing, and protection. The DMP aligns with EU data protection regulations and ensures that all project data are managed responsibly. The data management plan was made available to all consortium members for information and training purposes.

Q3-Q6. As the project involves sensitive groups of people and the transfer of personal data to third countries that do not have European data protection standards, the data management plan was supplemented by a joint controllership agreement. The main content was developed jointly with all consortium members, in particular with the data protection officer of the TUD and the legal advisor of the European Project Centre of the TUD. The joint controller agreement specifies individual and joint responsibilities in data processing and was submitted to the consortium for signature. It serves to provide legal protection for data processing and to raise awareness within the consortium. This was included within the *Deliverable D1.3. 1st update of Data Management Plan*.

Deviations and Impact on Other Tasks or Work Packages:

Deliverable D1.2 was resubmitted following the incorporation of feedback from CINEA. The required revisions were minor and, as a result, the adjusted timeline did not have any negative impact on subsequent tasks or work packages.

Task 1.4: Working Network Board formation and operation

M1-M48; TUD (WUST, all partners)

Achievements:

Q1-Q2. The foundations for establishing the Network of Eternal Partners (NEP) have been laid. The NEP is presented on the U_CAN project platform.

Q3. A digital application process for NEP membership was made available via the project website. The NEP was established as a central point of contact for external individuals and organisations. Members originate from several European countries as well as Ukraine. Sectors currently represented include municipal institutions, NGOs, research institutions and private companies.

Q4-Q6. Invitations were disseminated on a regular basis, and collaboration was expanded through the engagement and onboarding of additional relevant organisations. The partner map on the project platform was continuously updated. By the end of the reporting period, the NEP consisted of 18 members.

Deviations and Impact on Other Tasks or Work Packages:

Minor delay of *Milestone MS01. Working Network Partners Board established* due to late confirmation of some network partners and development of the interactive map of network partners for the U_CAN website. This had no impact on subsequent tasks or work packages.

1.2.2. Work Package 2: Creation of U_CAN project platform for knowledge sharing and networking (M1-M48, OMB)

WP2 Objectives

The aim of this WP is to establish and operate a project platform for data and knowledge sharing, and for stakeholder networking. The objective is:

- Establish a project platform for collaboration, engagement of stakeholder network and knowledge sharing among Ukrainian cities that are committed to the goal of carbon neutrality.

Progress in WP2 during the first Reporting Period of the project

WP2 focused on the development of the U_CAN Project Platform as a central tool for dissemination, communication, and stakeholder engagement. The platform was established to support project visibility and knowledge exchange and includes three interactive maps presenting the Network of External Partners, “Climate Neutrality”-related initiatives in Ukraine, and the suitability of locations for solar power station deployment. By the end of the reporting period, the general number of views at the Platform is 11 thousand with 548 new users and 563 active users. The engagement rate for the Platform is 62.9% which is considered successful for the platforms developed within EU projects.

Table 2: Overview of deliverables (D) & milestones (MS) of WP2

#	Title	Due date	Actual delivery	Delay justification
D2.1	Interactive map integrated with Project platform	30/11/2024	02/01/2025 (20/01/2025)	Integration of CINEA feedback following 1st submission
MS2	U_CAN Project platform established	31/10/2024	22/08/2024	n/a

Task 2.1: U_CAN project platform creation

M1-M6; OMB (TPF, ECSA, ENV, all partners)

Achievements:

Q1-Q2. The U_CAN project platform available at <https://ucan-ukraine.eu> was successfully developed and launched as a central knowledge-sharing and networking hub, facilitating stakeholder engagement, dissemination of best practices, and collaboration among key actors in climate neutrality. The platform design and development were undertaken in a phased manner, with initial structuring and content planning carried out in M1-M3, followed by technical implementation and content integration in M3-M4. The platform was launched in M4, with a subsequent review by all partners (Google forms were developed for submitting information about partners and pilot cities <https://forms.gle/AM5U1HoNruLQgMp38> <https://forms.gle/FroD3GTNStfMkcDq6>), final testing and deployment in M5-M6. All remarks, suggestions and corrections were duly implemented. TPF was responsible for the technical development, while OMB led the content structuring.

The platform operates as a bilingual (English/Ukrainian) resource to ensure accessibility for a broad range of stakeholders, including municipalities, policymakers, researchers, and industry representatives. Key functionalities include a dedicated repository for project results, knowledge and materials, event updates, news. The platform was developed using a flexible, GDPR-compliant Content Management System (CMS) (WIX), ensuring adaptability and ease of maintenance. The architecture integrates user-friendly navigation, accessibility-compliant design (WCAG 2.1), and embedded data visualization features, supporting structured content dissemination.

Q3-Q4. Preparatory work was undertaken to enable future expansion, including the integration of Interactive Mapping tools, a Learning Management System (linked to WP6), and additional features required for presenting project outcomes. The U_CAN platform will serve as a long-term knowledge

hub, remaining operational beyond the project duration to support urban climate neutrality efforts in Ukraine.

Q3-Q6. Regular platform maintenance and content updates were carried out by OMB and TPF in collaboration with ECSA (WP7) and other partners. This included publishing news and events in line with the WP2–WP7 dissemination algorithm, updating the knowledge repository, maintaining interactive map data, and ensuring interoperability with external platforms, including NetZeroCities and the sister project SUN4Ukraine.

Figure 2: U_CAN Project Platform structure

Deviations and Impact on Other Tasks or Work Packages:

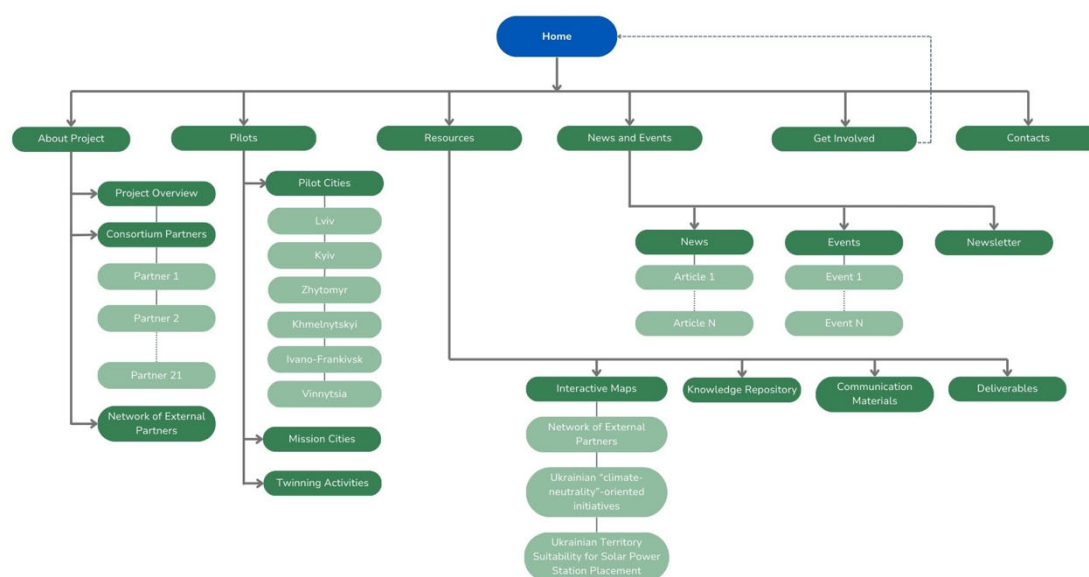
The U_CAN platform was established and launched two months ahead of the planned delivery date, resulting in a positive impact on the project by both supporting and accelerating the implementation of tasks, particularly within WP2 and WP7.

Task 2.2: Interactive Mapping tool

M7-M48; TPF (OMB, MCU, WUST)

Achievements:

Q1. The conceptual framework and functional scope of the U_CAN Interactive Mapping tools were defined, positioning the maps as key instruments for supporting collaboration among climate neutrality stakeholders in Ukraine and the EU, as well as for providing structured access to climate-



related data generated within the project.

Q2-Q6. Three key interactive maps (<https://www.ucan-ukraine.eu/interactive-maps>) were launched within this reporting period by TPF with the support of project partners:

- *Interactive Map of U_CAN External Partners Network:* This map visually represents the diverse array of organizations from Ukraine and the EU engaged in climate neutrality initiatives. It enables users to explore potential partners based on sector expertise, collaboration interests, and geographic location, thereby enhancing interdisciplinary cooperation.
- *Interactive Map of "Climate Neutrality"-Oriented Initiatives in Ukraine:* This map showcases ongoing and completed initiatives at municipal, regional, and national levels,

with filtering options by sector, project status, and location, facilitating strategic planning and matchmaking.

- *Interactive Map of Ukrainian Territory Suitability for Solar Power Stations Placement:* This map was developed to support the strategic planning and deployment of solar power infrastructure across Ukraine. Based on Deliverable D3.2 developed by ISKPI, the map applies a multi-criteria approach combining geospatial data, satellite observations, and analytical modelling to visualise territorial suitability.

All maps were developed using Power BI embedded analytics, selected for their advanced data visualization capabilities and dynamic interactivity. The interactive maps will continue to be expanded and refined in the upcoming project phases. Future developments include the integration of additional datasets to disseminate outcomes of Work Packages WP3-WP6.

Deviations and Impact on Other Tasks or Work Packages:

Deliverable D2.1 was submitted with a short delay, because data collection for the map has taken longer than expected and the internal review process was still implemented to ensure good quality deliverables. There was no impact on other deliverables or tasks.

1.2.3. Work Package 3: CNSC synergy building towards climate goals (M1-M48, WUST)

WP3 Objectives

The aim of this WP is to initiate and conduct twinning among the CNSC mission cities and other EU entities with collaboration-minded Ukrainian stakeholders involved in localised climate action initiatives. The objectives are:

- Develop consensus building in envisioning climate goals in various planning and reconstruction sectors in Ukraine and initiate teaming and partnerships between Ukrainian and CNSC Mission cities.
- Develop pilot Use cases for the Ukrainian cities of Lviv, Kyiv, Zhytomyr, Khmelnytskyi, Ivano Frankivsk and Vinnytsia that aspire climate neutrality.
- Support local authorities of pilot cities in building capacities for climate policy dialogues and creating opportunities for consensus-building with local communities and other stakeholder groups, including determining the economic and social benefits of the green transformation for all potential stakeholders and
- Collaborate with the Mission platform and identify online networking opportunities for collaboration with funding institutes, and additional stakeholders in interested Ukrainian cities with other EU organisations by network building to link Ukrainian cities with EU and UN climate goals.

Progress in WP3 during the first Reporting Period of the project

WP3 focused on identifying priority sectors for climate action in Ukraine and initiating structured knowledge exchange between Ukrainian pilot cities and EU Mission Cities. The work package delivered key analytical inputs on sectoral priorities and launched the twinning process through targeted exchange formats, laying the foundation for capacity building, pilot implementation, and long-term cooperation in support of climate neutrality objectives.

Table 3: Overview of deliverables (D) & milestones (MS) of WP3

#	Title	Due date	Actual delivery	Delay justification
D3.1	Base report: Sectors for Climate Actions in Ukraine	30/04/2025	30/07/2025	A 3-months extension was approved to improve the quality of the report with input from Ukrainian pilot cities during the consortium meeting in Wroclaw in Mai 2025
D3.2	Ukrainian Territory Suitability for Solar Power Station Placement	30/04/2025	05/05/2025	n/a
MS3	Collaboration with Mission Platform established	31/08/2024	06/03/2025	Signature of MoU with NetZeroCities has been delayed by negotiation of legal departments and receiving mandates from each U_CAN partner for TUD to sign the MoU
MS4	CNSC Synergy workshop conducted	30/11/2024	29/11/2024	n/a
MS5	Indicators for climate neutrality in the context of pilot activities	31/10/2025	31/10/2025	n/a

Task 3.1: Identification of key sectors in climate actions M4-M12; AU (ISKPI, WUST, ENV, JRG, SFPA)

Achievements:

Task 3.1 was planned for the first 12 months of the project and entailed (i) the identification of priority climate mitigation and adaptation sectors at the national and municipal levels and (ii) spatial assessment of renewable energy deployment conditions. Two submitted deliverables associated with the task are: (a) sectoral mapping of emissions and policy relevance (D3.1), and (b) geospatial suitability analysis for solar power investment siting (D3.2).

The task was led by AU, who was also the lead author of D3.1, with contributions and reviews from ENV, ULA, as well as WUST, JRG, and SFPA, while the principal authorship of D3.2 was held by ISKPI, with contributions and reviews from ZCC and JRG.

Q2-Q5. The *Deliverable D3.1. Sectors for Climate Actions in Ukraine* was based on document analysis of national climate legislation, policy frameworks, and local climate plans, complemented by a review of scientific literature and international best practice sources. Evidence was further informed through bilateral agreement reviews and targeted searches in multiple languages to capture non-English sources. Additionally, qualitative data were collected from six Ukrainian pilot cities through interviews, a workshop, and an open-ended questionnaire conducted in spring–summer 2025, offering insights into local priorities, barriers, and institutional capacities. This combination of desk research and stakeholder consultation enabled a multi-level assessment of climate action needs and sectoral opportunities in Ukraine.

According to the report, Ukraine's climate policy targets climate neutrality by 2050 and a 65% reduction in GHG emissions by 2030, framing green reconstruction as essential for EU integration and energy security. The energy sector is the core priority, alongside heating modernisation, renewable expansion, transport decarbonisation and waste management. In the six pilot cities, actions focus on heating upgrades, clean mobility, and resilience, and gain the strongest support when linked to energy security. Key barriers include limited financing, weak coordination and administrative capacity, and delays in the implementation of national instruments, all of which are amplified by wartime conditions. Despite this, strong city leadership, EU alignment, academic cooperation and public support create favourable conditions, with pilot initiatives expected to build capacity and demonstrate scalable solutions.

Q2-Q4. The *Deliverable D3.2. Ukrainian Territory Suitability for Solar Power Station Placement* employed an integrated, multi-criteria geospatial approach to evaluate land suitability for solar power stations across Ukraine. It utilised satellite-derived climatic variables (GHI, temperature, precipitation, wind), topographic datasets (elevation and slope), and a 2022 national land-cover map. All inputs were standardised through fuzzy logic to a 0–1 suitability scale. Their relative significance was determined using the Analytical Hierarchy Process and expert pairwise comparison, yielding consistent weighting coefficients. These weighted factors were then combined using the Weighted Sum Model to produce a national suitability index, with wetlands, rivers, and dense forests entirely excluded from potential zones. This process produced a spatially explicit suitability map that supports evidence-based identification of optimal solar PV deployment sites.

The geospatial assessment indicates that most of Ukraine's territory has medium to high suitability for solar PV development, with the highest values concentrated in the southern oblasts and Crimea, where suitability exceeds 0.75. Mountainous and densely forested areas, especially the Carpathians and northern Polissia, are mainly unsuitable due to steep terrain and low solar access. Existing large solar farms are mainly situated in zones classified as highly or very highly suitable, validating the model. Yet, several regions (including Poltava, Dnipropetrovsk, Donetsk, Luhansk, and northern Odesa) remain underexploited despite their high potential. Notably, 35.65% of land falls within the medium suitability class, 14.4% within the high suitability class, and 4.78% within the very high suitability class, revealing significant potential to expand decentralised PV systems to support national energy security and climate neutrality objectives.

The findings are now being used as the foundation for subsequent Tasks in WP3, mainly twinning activities, and to support alignment with WP4 pilot actions.

Deviations and Impact on Other Tasks or Work Packages:

The delivery of D3.1 was postponed from April to July 2025 due to restrictions on data collection from municipalities under wartime conditions. Limitations in administrative capacity and the lack of key technical staff required additional consultations and further interview rounds with Kyiv and Khmelnytskyi to complete the dataset. The revised timeline did not negatively impact subsequent Tasks or Work Packages.

Task 3.2: CNSC and Ukraine twinning initiation

M7-M30; WUST (LHD, TUD, TPF, all pilot cities in UA)

Achievements:

Task 3.2 commenced in November 2024 with the objective of initiating structured twinning between six Ukrainian pilot cities and Climate-Neutral and Smart Cities (CNSC) Mission Cities through the design and implementation of Knowledge and Experience Exchange Sessions (KEES, D3.3). The task builds on the analytical results of Task 3.1 and supports the implementation of Tasks 3.3 and 3.4.

Q2-Q3. A key element of the twinning process was the development of Knowledge and Experience Exchange Sessions (KEES) by WUST. These sessions were interactive formats designed to establish long-lasting, practically relevant collaboration between Mission Cities and Ukrainian partners. KEES aim to close capacity gaps, enhance governance standards, and facilitate the transfer of practical experiences in climate mitigation and adaptation.

KEES methodology structures twinning around: (1) needs-based sectoral analysis, (2) transferable solution identification, (3) local scenario development, (4) monitoring and policy alignment. The conceptual twinning structure is rooted in creating proximity (sectoral, organisational and geo-demographic) while building incentives for durable cooperation within the broader European decarbonisation agenda. Nine sectors were identified as key fields for exchange:

- construction and infrastructure,
- energy and utilities,
- industry,
- transport and mobility,
- waste and circular economy,
- environment and green infrastructure,
- spatial planning and land management,
- digital and Smart City solutions, and
- citizen engagement and social innovation.

Drawing on the identified sectors, a survey-based matchmaking process was completed to determine twinning constellations. Each Ukrainian pilot city was paired with two Mission Cities best aligned in terms of (1) sector-based expertise, and (2) geo-demographic and spatial comparability. The pairing logic is flexible, allowing iterative adjustments to secure the strongest possible synergies. However, based on the experience so far, the following matches are developing:

Figure 3: Initial twinning partnerships

TWINNING POTENTIAL: SECTORS

Pairs of cities	Construction and Infrastructure Sector	Energy and Utilities Sector	Industry Sector	Transportation and Mobility Sector	Waste Management and Circular Economy Sector	Environmental Management and Green Infrastructure Sector	Spatial Planning and Land Management Sector	Digital Solutions and Smart City Sector	Citizen Engagement and Social Innovation Sector
Ivano-Frankivsk-Košice		•		•					•
Ivano-Frankivsk-Wrocław		•		•			•		•
Khmelnitskyi-Bratislava					•				
Khmelnitskyi-Dresden		•		•	•				•
Kyiv-Košice		•		•			•		
Kyiv-Wrocław		•		•					
Lviv-Leipzig		•		•			•		
Lviv-Thessaloniki		•		•					
Vinnytsia-Bratislava		•					•		
Vinnytsia-Leipzig		•		•			•		
Zhytomyr-Dresden		•							
Zhytomyr-Thessaloniki	•	•		•					

Introductory survey results (November 2024)

Source: WUST research within the U_CAN Work Package 3.

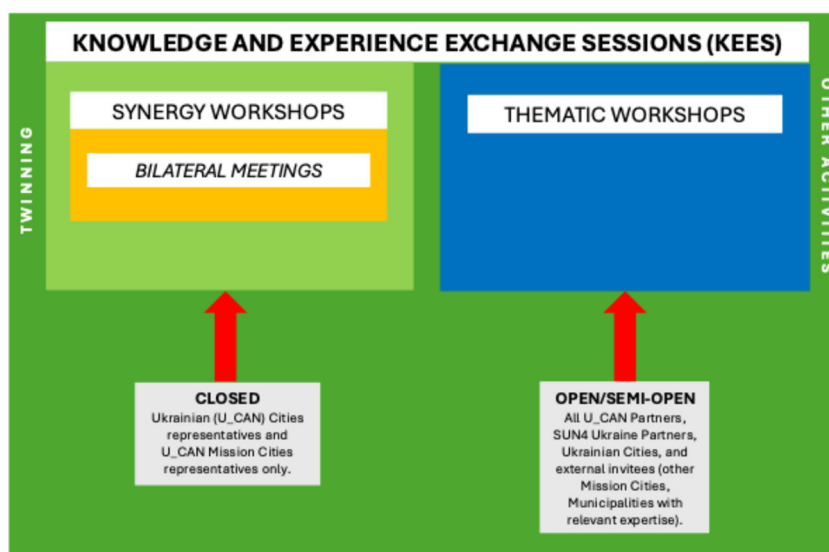
Q4-Q6: KEES activities were further divided into two streams as explained below:

(a) *Twinning-dedicated (closed)*: U_CAN Mission Cities + UA pilot cities – synergy workshops and bilateral meetings.

Synergy Workshops bring together U_CAN Mission Cities and Ukrainian pilot cities to build connections, exchange experiences, and identify shared priorities. Through guided discussions, participants define knowledge gaps, share good practices, and explore opportunities for mutual learning. The workshops help form a collaborative network, inspire new ideas, and lay the groundwork for future cooperation. They also support the selection of strong twinning partnerships by highlighting complementary strengths and areas for joint action. Within the first reporting period, two synergy workshops were organised by WUST – one online (Nov 2024) and one hybrid with the delegates from Mission Cities and UA pilot cities invited to Wrocław (May 2025).

Bilateral Meetings are focused one-to-one sessions between representatives of Ukrainian pilot cities and U_CAN Mission Cities. They address specific fields of expertise or explore the potential for twinning partnerships. These targeted conversations enable cities to discuss concrete needs, exchange specialised knowledge, and assess compatibility for future collaboration. The meetings help clarify expectations, deepen mutual understanding, and support the development of well-matched, purposeful partnerships within the U_CAN initiative. Over 20 bilateral meetings were facilitated by WUST till the end of the first reporting period, mainly online, but also including study visits.

Figure 4: KEES activities scheme



(b) Other knowledge exchange activities (open / semi-open): U_CAN partners, SUN4Ukraine partners, other Mission Cities external municipalities and experts – thematic workshops.

Thematic Workshops focus on the priority topics identified by Ukrainian U_CAN Cities. These targeted sessions enable more profound exploration of specific challenges, case studies, or fields of expertise. They serve as additional knowledge-exchange events, bringing together practitioners, experts, and researchers to provide specialised input and evidence-based insights. Three online Thematic Workshops were organised, involving Mission Cities, Ukrainian partners, SUN4Ukraine members, and selected external cities and experts:

1. Sectors for Climate Action in Ukraine (February 2025) - Led by AU, focusing on differentiated sectoral pathways.
2. Advancing Decarbonisation Through Solar and Low-Carbon Solutions (April 2025 - Led by LHD with contributions from Klagenfurt am Wörthersee and U_CAN partners, covering solar potential, land analysis, and district-scale decarbonisation models.
3. Public Engagement for Climate Neutrality (November 2025) - Košice and Wrocław shared governance and citizen engagement strategies, formats and measurement methods, accompanied by input from AU and WUST.

Deviations and Impact on Other Tasks or Work Packages:

None.

Task 3.3: Pilot use cases conception

M7-24; AU (WUST, LHD, TUD, TPF, AU, all pilot cities in UA)

Achievements:

Q3-Q4. Task 3.3 aims to translate the analytical outputs of Task 3.2 into city-specific transition scenarios and pilot use cases for five Ukrainian cities. It operationalises strategic climate and energy targets into concrete pathways and governance models, tailored to local institutional capacity.

Q5-Q6: AU, together with other WP3 partners and the pilot cities, is implementing the following ongoing activities:

- Creation of transition scenarios for each pilot city, outlining decarbonisation priorities, heating system modernisation options, PV integration, transport reforms, and climate-resilient reconstruction approaches.

- Identification of institutional pathways, specifying which actors (mayors, utilities, CSOs, housing providers, EU twinning partners) must coordinate to unlock energy transition steps and secure capacity for implementation.
- Assessment of localised renewable energy potential and post-war feasibility of decentralised energy solutions, drawing directly on Task 3.2 suitability results and climate-neutral governance frameworks.
- Development of a best-practice catalogue (D3.4) that compiles scalable and replicable models for Ukrainian urban transition, aligned with EU Mission Cities experience, IPCC guidance, and EGD principles.
- Linking transition options to lived realities of the cities, including institutional stability, displacement patterns, damaged heat networks, and needs for quick-deployment renewables (e.g. rooftop and floating PV, modular heat networks).
- Integration with twinning and CNSC mentoring, ensuring that pilot use cases benefit from structured exchange and not only technical recommendations.

Deviations and Impact on Other Tasks or Work Packages:

None.

Task 3.4: Engagement with CNSC Mission Platform

M1-48; WUST (LHD, TPF, TUD, ENV)

Achievements:

Task 3.4 is focused on building structured engagement between U_CAN partners and the CNSC Mission Platform, strengthening the twinning architecture between EU Mission Cities and Ukrainian pilot cities throughout the project period.

Q1-Q2: Coordination was initiated with key European climate-neutrality networks, including NetZeroCities, CrAft, Eurocities and ICLEI, ensuring that Ukrainian stakeholders gain access to established transition methodologies, governance models, and financial instruments used by Mission Cities.

Q2-Q3: Onboarding into the CNSC ecosystem through the signing of a Memorandum of Understanding was formalised, enabling continuous cooperation and knowledge exchange channels.

Q4-Q5. A dedicated U_CAN group was established within the NetZeroCities portal and is currently being populated with pilot city data, climate strategies, and contact information. This represents a first step towards integrating Ukrainian local authorities into the same operational digital environment used by EU Mission Cities. The NetZeroCities platform is expected to serve as a long-term infrastructure for knowledge exchange, matchmaking, benchmarking, and capacity building between Mission Cities and Ukrainian cities. In parallel, a Collaboration Agreement between U_CAN and NetZeroCities was drafted and agreed upon by both parties, with signatures underway by the end of the reporting period.

Q6. Engagement and collaboration with other complementary European and Ukrainian initiatives (DREAMER, ConnectED, Eurocities, ICLEI, EECU) was initiated. This helps in broadening the U_CAN collaboration network beyond twinning alone and aligning with European best practices on just transition, multilevel governance, climate finance, and post-war green reconstruction.

Deviations and Impact on Other Tasks or Work Packages:

Due to a delay in the formalisation of the Memorandum of Understanding between NetZeroCities and the U_CAN project, MS3 was achieved later than scheduled. No impacts on other tasks or work packages were observed.

1.2.4. Work Package 4: Pilot Implementation in Ukraine, supported by European front-runners (M13-M45, CIL)

WP4 Objectives

The aim of this WP is to implement short to medium-term climate mitigation strategies in 6 pilot cities in Ukraine (Lviv, Kyiv, Zhytomyr, Khmelnytskyi, Ivano Frankivsk and Vinnytsia). The objectives are:

- Support implementation of pilot activities in six cities to demonstrate climate-neutral city solutions and
- Integrate learnings from project and contemporary practices in short and medium-term strategies of local governing bodies in Ukraine.

Progress in WP4 during the first Reporting Period of the project

WP4 focused on establishing the methodological and operational foundations for pilot implementation, including the development of climate indicators, preparatory assessments, and early pilot-specific activities, thereby enabling the transition from planning and knowledge exchange to concrete, city-level implementation.

Table 4: Overview of deliverables (D) & milestones (MS) of WP4

Deliverable number	Deliverable title	Delivery date	Comment
No deliverables until M18			

#	Title	Due date	Actual delivery	Delay justification
MS5	Indicators for climate neutrality in the context of pilot activities	31/10/2025	31/10/2025	n/a

Task 4.1: Assessment of climate indicators in pilot cities

M13-M21; OMB (ENV, BRIT, all pilot partners)

Achievements:

Q4. Climate indicators to assess the progress in Pilot Activities were selected. The framework methodology was developed by OMB and ENV, implementing a complex approach uniting several methodologies of evaluating sustainability of cities and territories, including the UNESCAP methodology for National Sustainable Development Goals adapted for the purposes of the U_CAN project. During the first stage of the climate indicators development, the preliminary complex list of climate neutrality indicators sorted by the field of origin (e.g., Transport, Energy, etc.) was formed.

Q5. During the second stage the given indicators were evaluated by the representatives of the Pilot Cities by their relevance to their Pilot Use Cases and the availability of data. The latter states a particular challenge due to the martial law in Ukraine which states restricted access to a number of indicators, and the absence of centralized data collection systems, as well as the absence of data on certain climate indicators in Ukrainian municipalities. The preliminary selection of climate indicator groups was conducted at the Workshop within the Consortium Meeting in Wroclaw in M13.

Q6. During the third stage the collection of data on the selected climate indicators from the Pilot Cities was conducted via the protected file sharing system used in the project. A questionnaire was developed by OMB to gather additional information on sustainability in the Pilot municipalities. During M16-M17 six meetings were held between OMB, ENBV and the representatives of the Pilot Cities within the Project Consortium, including the representatives of the municipalities where the Cities are represented by the university partners in the Consortium. Milestone *MS5 Indicators for*

climate neutrality in the context of pilot activities was achieved by the end of M18 which marked the end of selection of the climate indicators.

The *Deliverable D4.1. Climate neutrality and impact indicators in Pilots* is being developed and will be submitted by the end of M21. The methodology given in the Deliverable will serve not only for the assessment of the Pilot Activities implementation but also will allow to indicate the key hazards and potential risks within the Activities. The scalability of the methodology will allow for disseminating the findings of the Task to the climate neutrality actions outside the U_CAN project.

Deviations and Impact on Other Tasks or Work Packages:

None.

Task 4.2: Climate-neutral pilot activities in Lviv

M13-M42; CIL (OMB, ENV)

Achievements:

As part of the U-CAN project, the city of Lviv is implementing two pilot initiatives:

(a) the Creation and Implementation of the Potential Energy District (PED)

Q5-Q6. CIL team focused on data collection to support the formation of the PED. Expert assistance was provided to develop a heat-supply scheme for the cities of Lviv. Based on the 2023 SECAP, Lviv aims to become climate neutral by 2050, with the energy sector and the creation of the PED playing a key role in achieving this goal.

- As part of the implementation, a one-stop-shop concept was developed. The primary objective of this office is to communicate with residents, providing information on how to form an energy-positive district within specific locations.
- In addition, preparatory activities for developing the municipal energy plan within the pilot project were carried out. The city currently faces a significant challenge in gathering relevant data for the energy sector. The plan will streamline data collection and support the continued development and implementation of the PED.

(b) the Formation of a European Integration Bureau and City Office in Brussels.

Q2-Q6. Initially, the pilot design anticipated “Creating a European Integration Bureau and an office in Brussels.” The European Integration Bureau was already actively operating at the beginning of the U_CAN project. Consequently, the pilot’s focus shifted more towards establishing an office in Brussels.

- During this period, we engaged with other cities that have implemented similar offices, such as Oslo and Izmir, to assess the feasibility and scope of full implementation.
- Another key idea that has emerged was the creation of a unique training program for municipal employees. This program would bring in experts and foreign specialists to familiarize local staff with EU legislation, inform them about grant programs currently relevant to Ukraine, and conduct training within programs that are presently closed to Ukraine, thereby facilitating Ukraine’s future deeper integration with the European Union once it becomes a full member.

Deviations and Impact on Other Tasks or Work Packages:

None.

Task 4.3: Climate-neutral pilot activities in Kyiv

M13-M42; ISKPI (OMB, ENV)

Achievements:

During the reporting period, the Kyiv pilot team developed the morphological analysis method, modified by the authors, and adapted to the complex problems of planning cities with climate neutrality goals, is among the most prospective ways of structuring complex problems and considering multiple influence criteria, and can be viewed as a part of an important process of transforming the scientific methods into management decisions.

Q4-Q5. The authors tested 8 automobile tunnels provided for by the General Plan of Kyiv. The obtained car tunnel prioritization for Kyiv city was a decision with several groups of risk (logistical, ecological, safety) with different approaches to form rankings, that enhances management capabilities. The morphological model allowed to assess the impact of 10 structural and functional factors, and process ca. 280 000 value configurations to obtain numerical estimates for tunnel rankings. The analysis of results reveals the leaders of the ranking, recommended for priority construction. Depending on the impact factor consideration, those are either tunnels 5, 4, 3 (approach 1), or tunnels 5, 7, 8 (approach 2), with tunnel 5 being the leader in both cases. Introduction of the climate neutrality factor to the set of the impact factors significantly influenced the tunnel prioritization results and provided more reliable results for the future development of Kyiv city. This approach could be an important factor in making management decisions for cities that are undergoing a transformation of the transport system on the basis of climate neutrality.

Q6. Requests for official data were submitted to Kyivmetropolitan and Kyivpastrans. The process of obtaining and verifying these data was completed and reflected in the team's interim report. Regular communication was maintained with project coordinators and other partner cities through monthly meetings to discuss progress, challenges, and next steps.

Deviations and Impact on Other Tasks or Work Packages:

The main deviation from the initial schedule concerns delay in obtaining official transport and environmental data from municipal utilities. Some datasets remain limited and require additional administrative permissions and requests, which has somewhat delayed the completion of certain analytical steps. However, this delay did not critically impact the overall progress of Task 4.3 or its integration with other Work Packages. The team compensated for this by conducting additional desk research, analysing publicly available data using verified open sources.

Task 4.4: Climate-neutral pilot activities in Zhytomyr

M13-M42; ZCC (OMB, ENV)

Achievements:

Within the U-CAN project, the city of Zhytomyr is developing two comprehensive plans:

(a) the study of the legal basis and technical characteristics of the construction of a floating solar power plant

Q4. The Zhytomyr team focused on a comprehensive and integrative approach to studying the legal basis for introducing solar energy on water in Ukraine. ZCC reviewed a number of Ukrainian regulatory acts that establish requirements for the construction and use of solar energy on water. It should be noted that, unlike in Europe, the use of floating solar power plants is innovative for Ukraine, which means that the topic has not been researched and there may be resistance from the community during implementation. Thus, the following basic regulatory and legal acts were studied: the Land Code of Ukraine, the Water Code of Ukraine, the Law of Ukraine 'On the Regulation of Urban Development,' the Law of Ukraine 'On the Basic Principles of State Climate Policy,' the Law of Ukraine 'On Alternative Energy Sources,' the Law of Ukraine 'On Strategic Environmental

Assessment,' the Law of Ukraine 'On Local Self-Government in Ukraine,' and the Law of Ukraine 'On Land Management.' The resolutions of the Cabinet of Ministers of 21 February 2025 No. 194 "On Approval of the Procedure for Organising and Conducting Public Hearings in the Process of Issuing an Integrated Environmental Permit (amendments thereto)" and the Order of the Cabinet of Ministers of Ukraine 'On Approval of the National Renewable Energy Action Plan for the Period until 2030 and the Plan of Measures for its Implementation" dated 13 August 2024 No. 761-r.

Q5-Q6. Within the framework of WP4 and WP7, the project was presented to the executive bodies of the council: the ecology department of the city council, the mayor and deputies, as well as the public utilities department of the city council, the municipal enterprise 'KATP-0628' of the Zhytomyr City Council, which will subsequently use the generated electricity. The legal model for the station's operation was presented in detail, tax obligations for land tax were studied, and the possibility of connecting (if necessary) the floating solar station to the central power grid was examined. As of 31 October 2025, the following results were achieved:

- Guarantees were received for the further implementation of the project from the city administration and the deputy corps. On 16 October 2025, we received an official decision from the city council granting permission to the Zhytomyr City Council's municipal institution, the City Development Agency, to develop documentation for obtaining a land plot under a water body for the further construction of a floating solar power station. This decision is binding in accordance with Article 122 of the Land Code of Ukraine. This is an extremely important decision, as it expresses the support of the city authorities for moving towards the development of alternative energy sources.
- Memorandum of cooperation was approved by the Zhytomyr City Council by decision No. 1497 of 16 October 2025, with Ocean Sun AS and Nicc LLC, which provides for the development, expansion and strengthening of relations between the parties regarding the creation and operation of renewable energy sources.
- An agreement with Leo Energy was established on preliminary arrangements for the construction of a floating solar power plant.
- The City Development Agency signed a contract with a contractor to carry out work on the formation and registration of the land plot on which the future floating power plant will be located.
- A memorandum was signed with the municipal enterprise KATP of the Zhytomyr City Council, which is responsible for the removal of household waste.

(b) a plan for the development of urban horticulture through the renovation of the city's greenhouse facilities.

Q4. ZCC sees this pilot as an opportunity to strengthen the community's food security, as a tool for improving communication and cooperation within the community, integrating internally displaced persons, a way to restore the trust of city residents in local authorities through the implementation of mechanisms for the temporary transfer of greenhouses to city residents for use. Thanks to the work in Work Package 3, we were able to find a partner city that has useful experience in the relevant field.

Q5-Q6. A comprehensive, informative and inspiring working trip to the city of Dresden was organised with a colleague from the Zelenbud Municipal Enterprise of the ZCC, which has greenhouses at its disposal and will be the legal entity that will subsequently transfer this property for use by citizens.

In order to renovate and develop the city's greenhouse industry, the ZCC will consider the example of shared gardens in Dresden and develop its own unique legal and practical model of cooperation between citizens and local authorities, based on the principles of transparency, cooperation and mutual responsibility, environmental friendliness, development and scaling. As of 31 October 2025, the following results were achieved:

- The project concept was presented and received positive feedback from the city administration regarding approaches to property use. We have conducted photo and video shooting of the greenhouse industry using a drone.
- The legal basis for the transfer of property for use was assessed for feasibility.
- The Land Code and the Laws of Ukraine: ‘On Farming’, ‘On Personal Peasant Farming’, ‘On the Lease of State and Municipal Property’ and the Civil Code of Ukraine in terms of easement rights was reviewed.
- The most important achievement, which is of fundamental importance for the further development of the pilot project, was the definition of the legal basis and model of cooperation between the ZCC and the greenhouse user, the definition of mutual rights and obligations, and the development of a preliminary cooperation agreement between the parties

Deviations and Impact on Other Tasks or Work Packages:

None.

Task 4.5: Climate-neutral pilot activities in Khmelnytskyi
M13-M42; KhNU (LYKUML, OMB, ENV, LHD)

Achievements:

Within the U-CAN project, the city of Khmelnytskyi is developing two pilots, led by two different university partners:

(a) Center for Local Policy Support in the Field of Climate-Neutral Construction and Energy, led by LYKUML

Q5-Q6. On October 24, 2025, on the International Day of Climate Action, the official opening of The Centre for the support of local policies in the field of climate- neutral construction and energy (hereinafter – the Centre) took place within the walls of Leonid Yuzkov Khmelnytskyi University of Management and Law. The event involved representatives of the city authorities, non-governmental organizations, business representatives, academic community, international partners, representatives of the U_CAN pilot partner cities, and the public.

The positive impact of the event was the increased level of public interest in issues of energy transformation and climate neutrality, the strengthening of partnership interaction between authorities, science, and the public, and the formation of a positive image of Khmelnytskyi as a modern, progressive and environmentally responsible city.

The establishment of the Centre is a logical step in the implementation of the city’s environmental strategy, as it will become a platform for scientific research, education, public initiatives, and international partnership in the field of climate-neutral development and energy. The Centre’s main objective is to conduct educational-scientific, and cultural-awareness activities in the field of developing climate neutrality, ecological, and energy transformation through the study of local policies, programmes, and practices, international cooperation, and the search for innovative proposals.

The Centre will ensure:

- The creation of a platform for dialogue and cooperation between authorities, businesses, non-governmental organizations, educational institutions, and scientists.
- Raising community awareness regarding contemporary environmental challenges and opportunities for sustainable development.
- Development of practical solutions in the fields of energy efficiency, waste management, and climate change adaptation.
- Support for the formation of local policies that will contribute to achieving climate neutrality and the sustainable development goals of the city of Khmelnytskyi.
- Training of specialists through workshops, experience exchanges, and educational initiatives. Identification of the target groups of the Centre’s for the support of local policies in

the field of climate-neutral construction and energy activities, including public authorities, civil society organizations, the business community, 2 educational institutions of Khmelnytskyi, as well as experts and researchers in the fields of ecology, climate neutrality and energy.

(b) Adaptive Traffic Management system to improve urban mobility and reduce CO₂ emissions in Khmelnytskyi, led by KhNU

Q5-Q6. Key activities completed during the reporting period include:

- An Adaptive Traffic Management System was introduced, designed to enhance the climate resilience of Ukrainian cities. The system integrates AI algorithms, smart sensors, and geo-spatial data to optimise traffic flows and support sustainable urban mobility.
- The article “An Adaptive Machine Learning Approach to Sustainable Traffic Planning: High-Fidelity Pattern Recognition in Smart Transportation Systems” — authored by Vitaliy Pavlyshyn, Eduard Manziuk, Oleksander Barmak, Pavlo Radiuk, and Iurii Krak — was published in the (Gold Open Access, SJR: Q2) MDPI journal Future Transportation (Basel, Switzerland), <https://doi.org/10.3390/futuretransp5040152>.
- A meeting was held with representatives of the Khmelnytskyi City Council and the Deputy Mayor to discuss collaboration on the implementation of scientific solutions developed under the U_CAN project. As a result, a working relationship was established with the Department of Transport Infrastructure to support the integration of adaptive control software into the city’s traffic system, advancing the city’s transition toward sustainable mobility.
- Preparations began for the KhNU Conference on CO₂ Emissions Reduction 2026 (MS6), including organisational arrangements and programme development to ensure effective dissemination of project results and exchange of expertise.

Deviations and Impact on Other Tasks or Work Packages:

None.

Task 4.6: Climate-neutral pilot activities in Ivano-Frankivsk
M13-M42; BRIT (OMB, ENV)

Achievements:

BRIT is responsible for the implementation of a photovoltaic (PV) installation within a selected public building located in the central part of Ivano-Frankivsk.

The final selection of the building will be based on the outcomes of a comprehensive needs assessment, taking into account dissemination potential, synergies with other ongoing or planned initiatives, and alignment with the overall project objectives. The planned installation will comprise a PV-panel solution, smart metering components, LED lighting, and ensure public accessibility for demonstration purposes.

Q3. BRIT established and maintained close cooperation with the Ivano-Frankivsk City Administration, in particular with Ms. Nadiia Kromkach, Deputy Head of the Department of Economic Development, Ecology, and Energy Saving of the Ivano-Frankivsk City Council.

Q4. During the application stage, the initially proposed costs for PV equipment (EUR 10,000) were removed, as U_CAN is classified as a soft project without direct hardware investment components. Nevertheless, BRIT identified an alternative approach by utilizing existing PV panels available for public buildings, thereby ensuring the continuation of this demonstrative activity within the project's framework. IT development services were used to establish an online platform to present, monitor, and promote the initiative.

Q5-Q6. In this regard, the BRIT team held a series of coordination meetings with representatives of the company responsible for the development and maintenance of the official website of the City of Ivano-Frankivsk. The collaboration will ensure the visibility, transparency, and replicability of the implemented actions in line with EU project dissemination and communication standards. By the end of the reporting period, the online platform was launched and disseminated through U_CAN channels.

BRIT's experts, in cooperation with representatives of the Ivano-Frankivsk City Administration, conducted joint work aimed at collecting and analysing data necessary for the development and completion of climate indicators. This collaboration contributed to enhancing the city's capacity for systematic climate monitoring and data-driven policy planning.

Deviations and Impact on Other Tasks or Work Packages:

None.

Task 4.8: Performance assessment of initiatives in six pilot cities

M22-M45; ENV (OMB, BRIT, all pilot partners)

Activity has not started yet.

1.2.5. Work Package 5: Modernization and standardisation of technologies and practices (M4-M45, ULA)

WP5 Objectives

The aim of this WP is to support the process of standardisation and modernisation of technologies. Objectives are:

- Standardisation activities relevant to the EU and international levels.
- Modernisation of green technologies in the building sector.
- Procedures and protocols development for building stocks and its reconstruction and
- Citizen participation and engagement in local climate mitigation strategies.

Progress in WP5 during the first Reporting Period of the project

As a first action within the WP5 activities, a survey was prepared to enable the pilot cities to express their development and reconstruction needs and interests with a view to carbon neutrality. In order to make the process effectively shared and feasible, the following actions were carried out:

- involvement of project partners who share the objective of modernising and standardising technologies and practices.
- identification of hot topics related to the carbon neutrality of the built environment.
- definition of the queries within each topic by the partners previously involved.
- collecting the questions, skimming and systematising them to avoid repetitions and overlaps that could lead to ambiguities.
- defining the survey structure.
- administration and sharing of the survey to the Pilot cities involved in the U_CAN project.
- analysis of results.

The survey results were presented at the 2025 Cities Mission Conference held on May 6-8 in Vilnius, Lithuania, and at the 2025 IEEE International Workshop on Metrology for Living Environment, session no. 9, "Modernization of technologies and practices towards cities carbon neutrality (U_CAN Project)," held on June 11-13 in Venice, Italy.

The results provided the basis for the Co-Creation Labs held in Wroclaw during the project's Consortium Meetings, which were then further explored in bilateral meetings with the Pilot Cities held in June and July 2025, during which it was possible to establish the cities' priorities and objectives.

Table 5: Overview of deliverables & milestones of WP5

#	Title	Due date	Actual delivery	Delay justification
D5.1	Overview of standardization landscape	30/04/2025	05/05/2025	n/a

Task 5.1: Standardisation activities

M4-M45; DIN (JRG, ULA, TUD, all pilot partners)

Achievements:

Q2. The standardisation landscape analysis was carried out. The process began with the collection of keywords by DIN and direct input provided by other WP5 partners, which were used to identify relevant standards. Initial research resulted in approximately 4,000 standards. The raw dataset was then cleaned by removing irrelevant results as well as standards that did not meet the criteria of the International Classification of Standards (ICS), reducing the dataset to approximately 1,600 standards. Keywords were clustered, these clusters were assigned to partners based on their areas of expertise. Partners evaluated the standards within their assigned clusters using a provided template and

predefined rating criteria. After the first rating process, the dataset was further refined to approximately 800 standards.

Q3-Q4. A second rating process was completed by all WP5 partners in February 2025. The *Deliverable D5.1. Overview of standardization landscape* was completed by end of the first year of the U_CAN project. This included an introduction to standardization, basic concepts, the main types of standardization documents, and a summary of the standards relevant to U_CAN

In parallel, a liaison with CEN/TC 465, “Sustainable and Smart Cities and Communities,” was approved and initiated by TUD. The final assignment of the liaison was completed and formalised, marking a significant step forward in aligning with relevant standardisation bodies. TUD participated in regular monthly Ad-hoc meetings of the working group.

Q5-Q6. Preparations were undertaken by DIN for the Standardization Potentials and Opportunities Workshop. As part of this preparation phase, two alternative 30-minute introductory sessions were offered on September 30th 2025 and October 2nd 2025, allowing consortium members to attend the session that best suited their schedule. These sessions provided the necessary background ahead of the main workshop. The main workshop was scheduled for November 3rd, with the goal of gathering insights on the priority areas of each pilot city and exploring how these can translate into potential standardization activities. It also aims to foster dialogue between pilot cities and experts and to identify ways in which these local opportunities may contribute to the broader standardization landscape.

Deviations and Impact on Other Tasks or Work Packages:

None.

Task 5.2: Modernisation of technologies for decarbonisation of industries

M10-M33; JRG (AU, ULA, OMB)

Achievements:

During this reporting period, the following activities were carried out:

- Identification of the needs of Ukrainian cities regarding decarbonisation measures.
- Determining the sources of CO2 emissions in Ukraine.
- Evaluation of the potential of Ukrainian cities to implement renewable energy sources.
- Economic assessment of specific renewable energy generation technologies.

Q4. In April 2025, WUST and ISKPI organized the Thematic workshop “*Synergy Building towards Climate Goals*“, with JRG taking an active part to gather and discuss insights on recent developments in solar power generation for urban infrastructure.

Q5. During the consortium meeting in May 2025, a further workshop was held to conduct an initial assessment and to identify decarbonisation areas and solution approaches relevant for the pilot cities. Together with all six pilot cities, their needs for various decarbonisation technologies were identified and prioritised.

In parallel, an economic assessment of the potential, production technologies, and supply options for exporting hydrogen from UKRAINE to the EU was contacted (Terekhov, Y., Tuerk, A., Blaettner, D., (2024) Options of exporting hydrogen from Ukraine to the EU. International Multi-disciplinary Scientific GeoConference: SGEM, Sofia, Vol. 4, Issue 2. <https://doi.org/10.5593/sgem2024v/4.2/s16.04.>).

Q5-Q6. A separate study by JRG analysed the prospects for the development of renewable energy in Ukraine and provided an assessment of investments in different renewable energy generation technologies. The results of this work were presented in (Terekhov, Y. (2025) Recovery of the energy system of Ukraine towards the development of renewable energy. BIS Energy and Engineering, Vol. 2, <https://unimma.press/conference/index.php/biseeng/article/view/361>).

In continuation of this work, opportunities for implementing decarbonization technologies using solar energy in urban infrastructure were identified by JRG (Terekhov, Y. (2025) Current Decarbonization Technologies in Ukrainian Cities in the Energy Production and Consumption Sector Using Solar Energy, 2025 IEEE International Workshop on Metrology for Living Environment (MetroLivEnv), Venice, Italy, 2025, DOI: 10.1109/MetroLivEnv64961.2025.11107025).

Deviations and Impact on Other Tasks or Work Packages:

None.

Task 5.3: Procedures for repurposing war-affected demolition materials
M10-M33; ULA (RCSD, OMB, ENV)

Achievements:

Task 5.3 focuses on three main areas:

- Debris management, with particular emphasis on selective demolition.
- Recovery and reuse of building components.
- Recycling of materials from decommissioned buildings.

Q4. A dedicated, iterative methodology was developed by ULA collaboratively with all task partners and refined based on feedback from pilot cities and the preparatory Ukraine Reconstruction Conference (Rome, July 2025). Each phase was revisited and adjusted according to the outcomes of subsequent steps, allowing continuous optimisation of analyses, decision-making, and intervention strategies. The methodology comprised seven interrelated phases:

- Characterization of typical buildings in the Ukrainian building stock, focusing on construction techniques and materials.
- Parametric quantification of materials and building components per unit of surface area.
- Development of an algorithm to identify representative building types through cartographic analysis.
- Statistical correlation of building types with material quantities.
- Selection of best practices and definition of strategies for selective demolition.
- Assessment of component reuse potential and recycling pathways for non-reusable materials.
- Development of a framework for building waste management, accounting for recoverable materials.

This iterative approach ensures continuous refinement of procedures and supports evidence-based strategies for the sustainable repurposing of materials from war-affected buildings.

Q5. Building on this, ULA conducted a detailed characterization of construction techniques and materials in typical buildings and developed a parametric table quantifying the prevalence of dominant materials and components (e.g., doors and windows, cladding, concrete panels). This work was carried out in synergy with task 5.4.

To gather detailed information on the typological, material, and construction characteristics of buildings, two questionnaires were distributed to pilot cities in close collaboration with task 5.4, and co-creation labs were held in July 2025 for more in-depth analysis, with the participation of interested partners. These co-creation labs built on those held in Wroclaw, during the Consortium Meeting. A comprehensive bibliographic review provided additional insights into Ukraine's architectural heritage.

Q6. The current integration phase focuses on implementing parametric calculation within a GIS environment, based on the methodology previously developed in spreadsheets. In collaboration with task partners, research was also focused on identifying best practices for selective demolition, component recovery, and material recycling. To advance these efforts, a thematic workshop is scheduled for 10 December 2025, aimed at refining strategies and fostering knowledge exchange on post-

disaster reconstruction and long-term urban resilience. The workshop, organized with the Special Agency for the Reconstruction of L'Aquila (USRA), will focus on „Rock, Excavated Soil, and Rubble in Reconstruction Following the 2009 Earthquake.“

Deviations and Impact on Other Tasks or Work Packages:

Although challenges arose in collecting information on rubble due to the war, a comprehensive bibliographic review has facilitated the reconstruction of the state of the art, enabling the development of an accurate and detailed understanding of the built heritage of many pilot cities. It also facilitated the advancement of task 5.4.

Task 5.4: Protocols for renovation/energy retrofit of existing buildings

M10-M36; ULA (OMB, RCSD)

Achievements:

During this reporting period, ULA advanced along five complementary strands:

(a) Adaptive reuse:

- **Q4.** The strategy of adaptive reuse focused on exploring the potential of Ukraine's building stock, particularly unused or underutilized structures, to address housing needs for displaced populations and vulnerable groups, including veterans affected by conflict-related injuries. A typological characterization was conducted in collaboration with task 5.3, emphasizing building layouts and adaptability rather than materials and construction.
- **Q5.** The goal was to identify temporary housing solutions and develop preliminary strategies for accessibility and inclusivity, in line with discussions from the Verona Conference (May 2025) and the Ukraine Reconstruction Conference (Rome, July 2025). To support this work, the same questionnaires previously described were distributed to pilot cities, and results were shared at the Cities Mission Conference 2025 in Vilnius (May 2025), ensuring alignment with local needs and ongoing collaborative activities.

(b) Energy efficiency:

- **Q4.** The research was organised into three areas of study: definition of the regulatory framework in light of European directives and Ukrainian regulations on energy efficiency; study of the construction and thermal characteristics of Ukrainian buildings through literature analysis (also in collaboration with the activities of task 5.3) and the aforementioned survey, in order to define the variables of the project's multi-criteria methodology; definition of possible case studies useful for validating the methodology.
- **Q5.** This last point was made possible thanks to the already cited bilateral co-creation labs, which identified Ivano-Frankivsk and Khmelnytskyi as the cities most interested in energy efficiency issues. The initial results of these joint development activities were also presented at the 2025 IEEE International Workshop on Metrology for Living Environment, session no. 9, "Modernization of technologies and practices towards cities carbon neutrality (U_CAN Project)," held on June 11-13 in Venice, Italy.

(c) Renewable solar energy:

- **Q4-Q5.** The following steps were taken: classification of building envelope construction systems; analysis of residential and specialized building types in the pilot cities; study of solar potential and production capacity per hectare of urban land based on recurring building types and settlement systems in the pilot cities.
- **Q6.** This territorial data was strategic for framing actions to introduce renewables into buildings according to a multi-level protocol currently being developed.

(d) Mitigation of urban floods:

- **Q4.** The hydraulics group was carried out: an initial phase of literature review regarding the state of the art of sewer systems and urban flood mitigation in Ukraine, a survey of the key climatological data needed for the hydrological characterization of the areas of each individual city using the ECMWF database (i.e., ERA5).
- **Q5.** The data are collected on a grid of points that fully covers each pilot city and were used to characterize extreme events; a study of the information currently available and freely downloadable from European portals concerning the topology of the water drainage network and the hydrological network of each pilot city.
- **Q6.** The objective is to systematically catalogue all hydrological information in order to integrate the work being carried out by the territorial planning group in relation to land-use transformation and nature-based solutions.

(e) Urban planning:

- **Q4-Q5.** Cutting-edge methodologies were being used to frame pilot cities, both from an urban-rural-natural interface criterion, as well as to provide insights into the urban areas. The techniques used refer to machine learning techniques, remote sensing data, and indicators engineering discipline, to provide a package of analyses capable of supporting sustainable planning.
- **Q6.** Additionally, the description of the natural features characterizing each study area is currently under development. To support sustainable land and urban planning, a set of ecosystem services values is being elaborated for several pilot cities, according to the necessities expressed by each of them. Future activities will envisage the use of the produced analyses and outputs to provide useful information to support land planning in each area.

Also in relation to the activities of this task, a thematic workshop is being organized entitled „The urban regeneration of smaller towns in Abruzzo after disaster: focusing on underground services“, in collaboration with the Special agency for the reconstruction of municipalities in the crater area (USRC) scheduled for November 27, 2025. The purpose of this event is to exchange knowledge and best practices on post-disaster reconstruction.

Deviations and Impact on Other Tasks or Work Packages:

None.

Task 5.5: Citizen participation in local climate governance

M13-M42; TUD (ECSA, SFPA, ENV, WUST, AU, ULA, UA-partners)

Achievements:

Q5. A preliminary in-person workshop was conducted during the consortium meeting in Wrocław in May 2025 to collect an initial overview of existing citizen engagement practices across the six Ukrainian pilot cities. Building on these insights, a structured assessment framework called the Ukrainian Readiness Index (URI). It was developed to evaluate the preparedness and institutional capacities of Ukrainian municipalities to engage in EU-funded climate and resilience initiatives. The URI data collection phase has been completed, and refinement of the data from the six municipalities is currently underway.

Q6. In parallel, a participatory methodology called Exploratory Scenario Planning (XSP) was designed to explore potential pathways for local climate action in post-war contexts. The first scenario mapping workshop under this framework was conducted in Dresden in September 2025, focusing on the Zhytomyr pilot case on urban gardening, with participation from local representatives, project partners and the city of Dresden. This workshop served as a pilot to test participatory formats for co-developing city-specific climate-neutral and recovery-oriented scenarios.

Deviations and Impact on Other Tasks or Work Packages:

No major deviations occurred. This has been no adverse impact on other tasks.

1.2.6. Work Package 6: Capacity building and localised knowledge mobilisation in Ukraine (M18-M45, ISKPI)

WP6 Objectives

The aim of this WP is to conduct training, technical capacity building for various target groups and conduct local exchanges in Ukraine. The objectives are:

- Technical capacity building via development of ad-hoc training programmes and
- Localised multi-sectoral exchanges in Ukraine.

Progress in WP6 during the first Reporting Period of the project

Within the Training and Technical Capacity Building work package, ISKPI carried out a series of activities aimed at strengthening technical competences, developing educational programs, and promoting local multi-sectoral exchanges in Ukraine.

- The project was presented at international scientific events under Science for Peace and Security (2024): the Advanced Research Workshop (ARW) supported by NATO (“Energy infrastructure resilience in response to war and other hazards”, September 23–26, 2024) and Global Energy, Environmental, and Cyber Security (“Capacity building of young scientist generation in energy security and climate neutrality”, October 8–16, 2024, Budapest University of Technology and Economics, Hungary).
- On April 23, 2025, a Memorandum of Understanding was signed at Igor Sikorsky Kyiv Polytechnic Institute between the Educational and Scientific Institute for Energy Saving and Energy Management (ES ISEM) and the Municipal Institution “Research Institute for Socio-Economic Urban Development”, formalizing their cooperation within the Horizon Europe U_CAN project.
- As part of educational and methodological activities, a new elective course “European Green Deal” was developed, and work on the Guide “Climate Neutrality of Cities” (D6.1) was initiated.

No deliverables or milestones were planned in WP6 for RP1.

Task 6.1: Training and capacity-building

M25-M36; ISKPI (TPF, ECSA, DIN, ENV, WUST, MCU)

Achievements:

Q5. WP6 partners (TPF, TUD, MCU) jointly developed and agreed on a coordination methodology aimed at ensuring the integrity and logical coherence of the guide “Climate Neutrality of Cities” - from the diagnostic stage to the formulation of measurable and financially viable actions.

Q6. A preliminary plan for the guide “Climate Neutrality of Cities” was also developed, along with an initial structural breakdown of its sections and the allocation of responsibilities among U_CAN project partners. ISKPI and TPF defined key coordination tasks, including the development of common templates and style guidelines for all chapters, the organization of short coordination meetings among section leaders to harmonize content and cross-references, and the joint preparation of the introductory and concluding sections presenting the guide as a single, logically structured pathway of U_CAN results. In parallel, ISKPI conducted a review of existing online training materials to assess their potential reuse within the U_CAN project; discussions on this matter among WP6 participants are ongoing.

Task 6.2: Climate Briefing for Mayors and city representatives of Ukraine

M18-M33; MCU (TUD, TPF)

Achievements:

Q5. MCU coordinated with external content providers to define and agree on the content of two training courses: the *Climate-Neutral Smart Cities* course and a *Leadership* course for mayors. In parallel, preparatory activities were undertaken to support course development.

Q6. Organisation and coordination activities were carried out in preparation for the mayors' pre-briefing session planned as part of the Annual Mayors' Summit in Kyiv in November 2025.

Task 6.3: Localised pilot-related knowledge mobilisation

M34-M45; ZCC (all pilot cities, TPF, BRIT)

Activity has not started yet.

Task 6.4: Political dialogues with Ukrainian decision-makers

M24-M42; ISKPI (TUD, TPF, MCU)

Activity has not started yet.

1.2.7. Work Package 7: Communication, dissemination and exploitation (M1-M48, ECSA)

WP7 Objectives

The aim of this WP is to ensure widespread communication, dissemination, and exploitation of U_CAN findings among the European and Ukrainian stakeholders and support the realisation of the EGD. It will also foster new collaborations towards climate-neutral practices in Ukrainian cities.

The objectives are:

- Communication of project activities, findings and events via the U_CAN project platform.
- Dissemination of project results, success stories and knowledge on various platforms and
- Exploitation of project results to foster further initiatives and collaborations.

Progress in WP7 during the first Reporting Period of the project

Table 6: Overview of deliverables & milestones of WP7

#	Title	Due date	Actual delivery	Delay justification
D7.1	Dissemination, exploitation and communication plan	31/10/2024	01/11/2024 (20/01/2025)	Integration of CINEA feedback following 1st submission
D7.2	1st update of the dissemination, exploitation and communication plan	31/10/2025	04/11/2025	n/a

Task 7.1: U_CAN communication and dissemination activities

M1-M48; ECSA (SFPA, all partners)

Achievements:

Over the past period, significant progress has been made in establishing a strong and recognisable presence through various communication channels and materials.

- **Q1-Q2.** Project visual identity has been developed, incorporating all imagery and graphic elements that represent U_CAN. This includes a defined set of colours, typography, templates (for Microsoft Word and PowerPoint), design components, images, and the project logo. This ensures a consistent and professional appearance across all communication materials, products, and services.

A dedicated project website was developed in collaboration with WP2 and officially launched in August 2024. This bilingual platform serves as the central hub for all project-related information and is regularly updated with news, activities, and project progress, ensuring public accessibility and engagement.

The project has established key social media channels to engage different target audiences:

- (a) Facebook (@U CAN Project), established in M3. – Primarily targeted at Ukrainian stakeholders, with Ukrainian as the main language and English as a secondary language. Gained 230 followers in 18 months.
- (b) LinkedIn (@U_CAN Project), established in M3. – Designed for EU audiences, managed in English to facilitate collaboration and knowledge exchange. Gained almost 400 followers in 18 months, and showed itself as project's primary communication channel.
- (c) YouTube (@U_CANProject). - This channel was established at the later stage (M17) in response to a higher than anticipated volume of video materials, following the decision to produce short videos that showcase the project's achievements at various stages.

Having a dedicated YouTube channel ensures wider reach, easier sharing, and better searchability, while providing a centralised repository where all project-related videos are systematically organised and publicly accessible. These videos are also embedded across relevant sections of the project's website to complement textual content and make reporting more visually engaging. Social media platforms will continue to serve as the primary channels for disseminating video updates to stakeholders.

- **Q3-Q4.** To enhance outreach efforts, WP7 lead has produced a general awareness-raising brochure, a postcard outlining U_CAN's primary aims in accessible language, postcards describing each pilot city's goals. Available in both English and Ukrainian, these dissemination materials serve as essential tools for conferences, webinars, and meetings, supporting effective project promotion.

The project's introductory video, the digital banners presenting each pilot city's objectives, and the roll-down banner – all developed by the WP7 lead – represent another set of important communication tools used across conferences, webinars, and meetings, thereby further strengthening effective dissemination and promotion of the project.

The project's first newsletter was issued in M12, distributed to subscribers, and published on the project website. Subscription is available via the website, and this option is regularly promoted through the project's social media channels.

These achievements provide a strong foundation for increasing U_CAN's visibility, engagement, and impact among key stakeholders and the broader public.

Two deliverables were prepared by the WP7 lead. Both deliverables are based on the dissemination and exploitation framework outlined in the project proposal and provide a detailed operational plan, including defined target audiences, priority events, published materials, and key performance indicators for the various communication and dissemination activities.

Deviations and Impact on Other Tasks or Work Packages:

Any initial delays in communication and dissemination in the early stages were addressed and improved, ensuring smoother coordination moving forward. Collecting input on communication and dissemination activities from all partners remains challenging. To facilitate the process, partners are regularly reminded to complete the general project management tables on the data repository with their activities and to use the media submission form provided by the WP7 lead, which is intended to ensure timely updating of project media channels.

Task 7.2: Consolidation of U_CAN success stories

M22-M45; TUD (pilot cities, all partners)

Activity has not started yet.

Task 7.3: Exchanges with Mayors of Europe and Global Covenant of Mayors

M31-M45; MCU (CIL, SFPA, LHD, TPF)

Activity has not started yet.

Task 7.4: Annual Central European Energy Conference

M22-M48; SFPA (RCSD, ECSA, all partners)

Originally, the SFPA activity within the project planned to support two larger-scale conferences (at regional level and with political presence), continuing the tradition of the Central European Energy Conference (CEEC) as a high-level platform for discussing regional policy solutions. However,

during 2025, and following consultations with regional partners and key stakeholders, SFPA redesigned the format of the event to be more practice-oriented and bottom-up. This change directly responded to the expressed needs of municipalities and local actors, who emphasised the importance of practical exchange, peer learning, and concrete implementation-oriented discussions rather than purely strategic or political debates.

Q6. In order to achieve this goal,

- SFPA organised the Central European Energy Conference in cooperation with the Union of Cities of Slovakia, with the two Mission Cities (Košice and Bratislava) actively involved as members. The Union of Cities of Slovakia represents an important player with significant regional impact and a wide network of municipal partners across Slovakia and Central Europe.
- In addition, SFPA closely cooperated with the Carpathian Platform and the Association of Energy Efficient Cities of Ukraine. These partners enabled the participation of mayors and city representatives from Ukrainian cities beyond the core U_CAN project partnership, allowing them to share expertise and practical inputs generated within the project and to connect them with experiences from U_CAN partners and also other Central European municipalities.
- Although the formal reporting period ended in October 2025 and the conference itself was held in November 2025, a substantial amount of preparatory work was carried out by SFPA and project partners during the reporting period. This preparation phase included the conceptual design of the conference, definition of its thematic focus in line with U_CAN objectives, identification and coordination of partners and speakers, and the development of the detailed programme structure. Significant effort was devoted to aligning the content with the needs of municipalities, securing the participation of Ukrainian and Central European cities, and ensuring balanced representation of policy, technical, research and governance perspectives.
- In addition, organisational and technical preparations were undertaken, including venue arrangements, technical equipment, interpretation from/to Ukrainian language, coordination with partner organisations and targeted outreach to municipalities and stakeholders. This preparatory process required close cooperation among project partners and external organisations and represented an important investment into building trust, strengthening networks and ensuring high-quality outcomes.

Q7. The conference was scheduled to take place on 24–25 November 2025.

- The conference achieved rich participation from Ukrainian partners, including representatives from Khotyn, Lozova, Sheptytskyi, Uzhhorod, Rivne, Lviv, Ivano-Frankivsk, Zhytomyr, Khmelnytskyi, Bar and Kovel. In parallel, a strong presence of cities and regional representatives from Central Europe was ensured, especially from Slovakia, Czechia and Poland. Importantly, the event also attracted participants outside the formal project network, which significantly increased its added value in terms of networking, knowledge dissemination and visibility of U_CAN project results. For example, representatives of the Austrian Union of Cities were present, contributing to cross-country exchange and reinforcing the transnational character of the conference.
- Alongside 4 strategic plenary sessions, the conference programme included a comprehensive set of thematic roundtable discussions focused on the concrete needs of cities and regions. These covered the following topics: solar energy in municipalities, EU budgetary tools for municipal recovery and decarbonisation, geothermal heating systems for urban use, climate governance and citizen participation, energy performance standards and local building codes, circular economy strategies at municipal and regional level and adaptation measures for cities.
- U_CAN project partners were actively involved as expert contributors, particularly in the discussions on climate governance and citizen participation, energy performance standards

and local building codes, and circular economy strategies. Experts from TUD, AU, ULA, JRG, CIL and ZCC contributed their thematic knowledge, empirical findings and practical experience, ensuring a strong link between project results and real-life municipal challenges. These sessions were explicitly designed to provide practical know-how, facilitate peer-to-peer learning, and foster direct cooperation between municipalities, including Ukrainian partners operating under challenging conditions.

For U_CAN project partners, the event served as a platform for presenting project results, validating findings through practitioner feedback, and strengthening the project's visibility and impact beyond the consortium. Overall, the conference contributed to long-term capacity building, network consolidation and the creation of foundations for future joint initiatives in the field of local climate and energy transition.

Task 7.5: Initiation of small-scale projects with donor funding

M31-M48; RCSd (SFPA, JRG)

Activity has not started yet.

Task 7.6: Roadmap to empower climate action collaboration

M34-M48; ECSA (JRG, CIL, AU, TPF, ENV)

Activity has not started yet.

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1.3 Impact

U_CAN presented five Activities envisioning Impacts towards the vision within the DoA:

(a) Mapping of Ukrainian cities → Long-term Ukrainian City matches for Climate initiatives

During the first reporting period, U_CAN successfully advanced the identification and engagement of Ukrainian cities committed to climate neutrality. This process confirmed cities' alignment with EU climate objectives and supported the identification of priority sectors requiring targeted support, thereby establishing a strong basis for meaningful participation in the Climate-Neutral and Smart Cities (CNSC) Mission.

- *Five pilot cities*—Lviv, Kyiv, Zhytomyr, Khmelnytskyi, and Ivano-Frankivsk—were identified as the core group. Relevant representatives from each city administration are formally engaged to implement the planned pilot activities and to advance their cities' long-term climate neutrality visions.
- *Partnerships were established with the City of Kharkiv*, which joined the Network of External Partners (NEP) and invited to key project workshops.
- *Collaboration was initiated with national and regional networks*, including the Energy Efficient Cities of Ukraine (EECU) association, providing access to a broader pool of municipalities. Through these networks, additional interested cities will be engaged as network partners in subsequent phases of the project to support the continuation and scaling of climate neutrality actions.

(b) Supporting Ukrainian cities in the sectors identified → Strategic sectoral empowerment of cities in UA

U_CAN strengthened strategic sectoral support for Ukrainian cities by expanding its engagement beyond consortium expertise and systematically building an extended expert ecosystem around the identified priority sectors.

- *The Network of External Partners (NEP)* was established and is being expanded to include additional European cities, sector experts, and organisations with relevant experience in climate-neutral transition, reconstruction, and sectoral innovation, providing Ukrainian cities with access to a broader pool of knowledge and practical support.
- *Liaison with CEN/TC 465, "Sustainable and Smart Cities and Communities,"* was established, strengthening alignment between Ukrainian city needs and European standardisation processes in key sectors related to sustainable transformation and climate neutrality.

(c) Twinning and teaming Ukraine with CNSC Mission cities → Long-term CNSC partnerships

As described in detail in the WP3 section, U_CAN advanced structured twinning and teaming between Ukrainian cities and CNSC Mission Cities. Twinning activities were operationalised through multiple interaction formats, including 3 Synergy Workshops and over 20 bilateral meetings. These collaborations enabled Ukrainian cities to access practical experience, tested solutions, and governance approaches from European counterparts, while also facilitating reciprocal knowledge exchange with Mission Cities.

(d) Facilitating Use case based Pilot initiatives → Feasible adaptation of climate actions in Ukraine

As outlined in the WP3 and WP4 descriptions, U_CAN initiated the development of use case-based pilot initiatives derived from twinning collaborations between Ukrainian cities and CNSC Mission

Cities. During the reporting period, preparatory activities focused on translating analytical outputs and twinning exchanges into localised pilot concepts tailored to the contexts of the six Ukrainian pilot cities (Lviv, Kyiv, Zhytomyr, Khmelnytskyi, Ivano-Frankivsk, and Vinnytsia). The pilot implementation activities will continue and intensify in the subsequent project period.

(e) Disseminating and Exploiting best practices, success stories and CNSC collaboration → Climate consciousness in various sectors of Ukraine and New avenues for integration of EGD in Ukraine

During the reporting period, dissemination and communication activities focused on establishing visibility, accessibility, and knowledge-sharing channels through the U_CAN project website, communication materials, events, and engagement with European and international networks, as detailed in the WP7 descriptions. These initial actions targeted city administrations, decision-makers, practitioners, and civil society, creating a foundation for broader climate consciousness across sectors. In parallel, U_CAN engaged with European and international initiatives and platforms, including NetZeroCities, extending the reach and visibility of Ukrainian cities and opening pathways for future integration with European Green Deal-aligned processes. While the systemic exploitation of best practices and large-scale uptake will intensify in later project phases, the groundwork for long-term dissemination, replication, and integration of climate-neutral approaches in Ukraine was established during the first reporting period.

1.4 Update of the plan for exploitation and dissemination of results

During the reporting period, the consortium maintained the core objectives and target groups defined in the original Plan for communication, dissemination and exploitation, while refining approaches in response to stakeholder feedback and practical experience. The main updates relate to stronger use of multimedia formats, bilingual dissemination materials tailored to Ukrainian and European audiences, and a more practice-oriented design of key events to better support municipal needs.

Enhanced multimedia and dissemination materials

The project expanded its use of video beyond what was initially foreseen, creating a dedicated YouTube channel for convenient storage, referencing and re-use of audiovisual content. In addition to general project clips, new videos were produced to showcase partners' impressions from events and to highlight their achievements in specific time frames, thereby supporting ongoing visibility and exploitation of project results.

The consortium also broadened its portfolio of visual and printed materials in response to the needs of pilot cities and stakeholders. Digital banners were developed to represent U_CAN pilot cities at conferences and other events; and additional postcards were produced: a general postcard explaining U_CAN's primary aims in accessible language plus city-specific postcards describing each pilot city's goals. Produced in both English and Ukrainian, these materials have become essential tools for conferences, webinars and meetings, enabling clear and targeted communication of project objectives and emerging results.

Central European Energy Conference: strategic adjustment

In the case of the Central European Energy Conference, the planned activity was adjusted from a predominantly high-level, policy-oriented format to a more practice-oriented and bottom-up event concept. This shift responded to feedback from municipalities and local actors, who expressed a need for implementation-focused discussions, peer learning and concrete exchange on local climate and energy solutions. The revised format strengthened the role of municipal networks and partners,



broadened participation beyond the core consortium and enhanced opportunities for networking, knowledge dissemination and future exploitation of U_CAN results.

Overall, these updates reinforce the exploitation potential of U_CAN results by improving their accessibility, relevance and practical uptake.

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2. FOLLOW-UP OF RECOMMENDATIONS AND COMMENTS FROM PREVIOUS REVIEW(S)

Not applicable.

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3. EXPLOITATION PRIMARILY IN NON-ASSOCIATED THIRD COUNTRIES

Not applicable.

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4. OPEN SCIENCE

The research activities within the project involved local stakeholders in the design of the pilot activities and were based on the local needs of the pilot municipalities and their respective stakeholders. Key findings were presented to the pilot municipalities in thematic workshops, and other European and Ukrainian stakeholders were also involved. The public results were compiled and will be included in the project platform for knowledge exchange and networking. The first data sets and results have been included in the EU's open research repository, Zenodo, among others. This means that various stakeholders have already contributed to the process and implementation of open science practices at an early stage, including the exchange of knowledge and contacts with U_CAN's sister project, SUN4Ukraine, and the Mission Platform.

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5. DEVIATIONS FROM ANNEX 1 AND ANNEX 2 (IF APPLICABLE)

All deliverables and milestones due by M18 have been submitted or achieved. Although some deliverables experienced a minor delay (up to three months in case of D3.1), there is no further impact on other deliverables or milestones foreseen.

5.1 Tasks/objectives

There are no deviations from planned tasks or objectives. Some tasks, e.g. in WP4 and WP6, have started a little earlier than planned with preparatory activities.

5.2 Use of resources

Overall, 25% of the budget has been spent during the first 18 months of the project, which is lower than the amount planned assuming a linear distribution. Costs are expected to increase in the 2nd and 3rd reporting period, thereby offsetting the lower expenditure observed in the 1st reporting period.

Table 7. Effort spent per partner and per work package (M1 – M18); linear distribution of planned effort assumed

P.no.	# months	WP1		WP2		WP3		WP4		WP5		WP6		WP7		Total		Deviation
		Plan	Actual	Plan	Actual	Plan	Actual	Plan	Actual	Plan	Actual	Plan	Actual	Plan	Actual	Plan	Actual	
		18	18	18	18	18	18	6	6	15	15	1	1	18	18	18	18	
1	TUD	15,00	7,37	0,00		3,00	0,73	0,00		2,50	0,27	0,11	0,00	6,75	0,66	27,36	9,03	-67%
2	LHD	1,13	0,82	0,00		3,00	2,68	0,36	0,31	0,00		0,00		3,00	1,88	7,49	5,69	-24%
3	ECSA	1,50	0,52	1,50	0,03	0,00		0,00		1,43	0,86	0,07	0,00	8,63	12,65	13,13	14,06	7%
4	DIN	0,38	0,26	0,00		0,00		0,00		3,57	4,13	0,04	0,00	0,38	0,03	4,36	4,42	1%
5	SFPA	0,75	0,25	0,00		0,38	1,00	0,00		0,36	0,20	0,00		9,75	0,38	11,23	1,83	-84%
6	ENV	0,75	0,68	0,38	0,95	1,88	2,92	2,73	3,98	2,14	2,84	0,04	0,28	2,25	0,80	10,16	12,45	23%
7	TPF	1,50	0,97	4,88	7,53	2,25	1,92	0,00		0,00		0,36	0,34	2,63	1,90	11,61	12,66	9%
8	WUST	3,00	3,08	1,13	0,88	7,88	9,91	0,00		0,36	0,21	0,07	0,08	1,13	0,63	13,55	14,79	9%
9	AU	0,75	0,84	0,00		4,50	5,97	0,00		1,07	0,31	0,00		1,13	0,14	7,45	7,26	-3%
10	ULA	3,00	1,67	0,00		0,00		0,00		10,36	3,60	0,00		2,25	0,28	15,61	5,55	-64%
11	JRG	0,75	0,86	0,00		1,13	1,69	0,00		3,21	4,33	0,00		4,50	0,00	9,59	6,88	-28%
12	BRIT	0,75	0,75	0,00		0,75	0,75	2,55	2,55	0,36	0,34	0,07	0,07	1,13	1,13	5,60	5,59	0%
13	OMB	0,75	0,75	4,50	8,17	0,38	1,00	3,27	2,01	1,43	0,00	0,00		1,88	0,00	12,20	11,93	-2%
14.1	ISKPI-EE	1,50	1,50	0,00		1,13	3,00	6,55	8,55	0,36	0,34	0,71	0,71	2,25	2,25	12,49	16,35	31%
14.2	ISKPI-MM	1,50	0,25			3,75	10,00	2,00						1,88	0,25	9,13	10,50	15%
15	MCU	0,75	0,80	1,50	1,70	0,00		0,00		0,00		0,43	1,50	4,50	5,00	7,18	9,00	25%
16	CIL	1,50	1,00	0,00		0,75	0,60	6,55	4,65	0,36	0,28	0,29	0,23	4,13	2,50	13,56	9,26	-32%
17	ZCC	0,75	0,36	0,00		0,75	1,00	3,64	3,53	0,36	0,34	0,21	0,00	1,88	0,86	7,58	6,09	-20%
18	KhNU	0,75	2,73	0,00		0,75	0,61	4,36	1,48	0,36	0,18	0,07	0,00	1,88	1,05	8,17	6,05	-26%
19	LYKUML	0,75	0,98	0,00		0,75	1,73	3,27	2,12	0,36	0,00	0,07	0,11	1,88	0,42	7,08	5,36	-24%
20	ME IUD	0,75	1,05	0,00		0,75	0,84	3,64	15,66	0,36	0,35	0,21	0,21	1,88	1,19	7,58	19,30	155%
21	RCSD	0,75	0,75	0,00		0,00		0,00		3,57	4,05	0,00		5,25	6,11	9,57	10,91	14%
	Total:	39,00	28,24	13,88	19,26	33,75	46,35	38,91	44,84	32,50	22,63	2,75	3,53	70,88	40,11	231,66	204,95	-12%
	Deviation	-27,6%		38,8%		37,3%		15,2%		-30,4%		28,4%		-43,4%		-11,5%		

Justification of effort deviations <20% at WP level during the 1st reporting period:

WP	Justification for deviation
WP1	Underspending is mainly due to the Coordinator TUD not being able to claim all actual costs due to missing documentation. This will be adjusted with the 2 nd Periodic Report.
WP2	WP2 Leaders TPF and OMB created and populated the U_CAN platform during the first months of the project, which required a lot of effort leading to overspending of person months compared to a linear distribution.
WP3	Overspending: WUST as WP-Leader had additional effort to coordinate the WP activities and ISKPI-MM used most of their effort planned for WP3 during the 1 st Reporting Period for the preparation of D3.2.
WP5	Underspending: Most tasks in WP5 only started in M10 and are ramping up, which led to underspending, mainly of the WP-Leader ULA, during the 1 st Periodic Report.
WP6	Overspending is in effect only 3 PM, mainly by ISKPI-EE due to the expanded scope of Training and Technical Capacity Building activities.
WP7	Overall there was an underspending of effort in WP7 because dissemination and exploitation activities have only started during the 1 st reporting period and will increase in the 2 nd half of the project.

Justification of effort deviations <20% at partner level during the 1st reporting period:

Partner	Justification for deviation
TUD	Lower actual effort reported due to missing documentation, which will be adjusted with the 2 nd Periodic Report.
SFPA	Lower effort mainly in WP7 because the CEEC conferences, for which SFPA is responsible, will only take place in the 2 nd and 3 rd reporting period.
ULA	Lower effort mainly in WP5, which is led by ULA. Most tasks only started in M10 and are ramping up leading to lower effort during the 1 st reporting period. In addition, recruitment of a young researcher hired specifically for the project has been delayed and the contract has not started yet.
JRG	Lower effort mainly in WP7, because dissemination and exploitation activities will take place mainly in the 2 nd and 3 rd reporting period.
ISKPI	Overspending
CIL	Lower effort mainly in WP7, because dissemination and exploitation activities will take place mainly in the 2 nd and 3 rd reporting period.
KhNU	In effect, effort is only 2PM lower than planned, mainly in WP4, where activities have only started at the end of the 1 st reporting period and will increase in the 2 nd reporting period.
ME IUD	Higher effort than planned mainly in WP4, where activities have started much earlier than planned to support the research work on the small river Tiazhyliv to understand the limits and framework for creating a socio-spatial concept for its development and a strategy for restoring its biodiversity.

Table 8. Planned versus actual costs (M1 – M18) - linear consumption of resources assumed – and planned versus actual person months rate (M1-M18)

Part. no.	Partner name	Planned Total costs	Actual Total costs	Deviation of costs	Planned PM-rate	Actual PM-rate	Deviation of PM-rate
1	TUD	299.062,50 €	88.668,95 €	-70%	7.000,00 €	6.810,76 €	-3%
2	LHD	65.390,63 €	46.242,48 €	-29%	6.000,00 €	6.135,51 €	2%
3	ECSA	126.703,13 €	109.484,51 €	-14%	6.400,00 €	5.874,65 €	-8%
4	DIN	51.796,88 €	40.913,80 €	-21%	7.500,00 €	7.154,79 €	-5%
5	SFPA	84.140,63 €	7.325,39 €	-91%	2.500,00 €	2.587,02 €	3%
6	ENV	74.895,38 €	84.169,93 €	12%	3.838,08 €	4.956,12 €	29%
7	TPF	135.937,50 €	118.688,15 €	-13%	6.600,00 €	7.033,14 €	7%
8	WUST	134.953,13 €	159.035,89 €	18%	6.300,00 €	7.395,56 €	17%
9	AU	89.250,00 €	80.263,08 €	-10%	8.200,00 €	8.502,89 €	4%
10	ULA	112.171,88 €	54.885,94 €	-51%	4.404,65 €	7.119,52 €	62%
11	JRG	69.046,88 €	59.683,15 €	-14%	2.000,00 €	6.430,15 €	222%
12	BRIT	30.000,00 €	17.301,94 €	-42%	2.000,00 €	1.997,70 €	0%
13	OMB	63.656,25 €	40.304,90 €	-37%	2.500,00 €	2.458,80 €	-2%
14.1	ISKPI	115.078,13 €	69.365,01 €	-40%	2.000,00 €	1.959,92 €	-2%
15	MCU	88.359,38 €	94.301,15 €	7%	2.666,67 €	2.298,05 €	-14%
16	CIL	53.671,88 €	26.408,25 €	-51%	2.000,00 €	2.000,28 €	0%
17	ZCC	53.671,88 €	23.338,75 €	-57%	2.000,00 €	1.926,27 €	-4%
18	KhNU	54.093,75 €	7.095,01 €	-87%	2.000,00 €	330,51 €	-83%
19	LYKUML	49.734,38 €	10.912,59 €	-78%	2.000,00 €	840,46 €	-58%
20	ME IUD	50.156,25 €	50.536,61 €	1%	2.000,00 €	1.041,19 €	-48%
21	RCSD	48.656,25 €	39.351,13 €	-19%	2.000,00 €	2.414,29 €	21%
	Total:	1.850.426,63 €	1.228.276,58 €	-34%			

Justification of deviations <20% of total costs during the 1st reporting period:

Partner	Justification for deviation
TUD	Underspending in personnel costs due to missing documentation, which will be adjusted with the 2 nd Periodic Report. Underspending in travel costs, since the kick-off meeting was held in Dresden and no travel costs were paid yet for network partners.
LHD	Underspending in personnel costs because of the late start of the contract for the person working on U_CAN. Initial contributions were covered by permanent staff without claiming costs.
SFPA	Underspending in personnel costs mainly in WP7, which is due to dissemination and exploitation activities starting later in the project, in particular the activities related to the CEEC conferences. This also explains the underspending in purchase costs as most of these costs are associated with the CEEC conferences, which only start in the 2nd reporting period.
ULA	Underspending in personnel costs mainly in WP5, which is led by ULA. Most tasks only started in M10 and are ramping up leading to lower effort and hence costs during the 1st reporting period. In addition, recruitment of a young researcher hired specifically for the project has been delayed and the contract has not started yet. Furthermore, the subcontract for WP5 has not been started yet.
BRIT	Underspending in personnel costs because the main activities in WP4 and WP6 will be in the 2 nd and 3 rd reporting period.
OMB	Underspending in personnel costs as most effort is planned for WP4, which only started at the end of the 1 st reporting period. Underspending in purchase costs as most costs are planned for dissemination activities occurring later in the project.
ISKPI	Underspending
CIL	Underspending due to falling exchange rate. Underspending in personnel costs as most effort is planned for WP4, which only started at the end of the 1 st reporting period.
ZCC	Underspending due to falling exchange rate. Underspending in personnel costs as most effort is planned for WP4, which only started at the end of the 1 st reporting period.
KhNU	Underspending due to falling exchange rate. Underspending in personnel costs as most effort is planned for WP4, which only started at the end of the 1 st reporting period and due to lower average monthly personnel costs than planned.
LYKUML	Underspending due to falling exchange rate. Underspending in personnel costs as most effort is planned for WP4, which only started at the end of the 1 st reporting period and due to lower average monthly personnel costs than planned.

Justification of deviations $\leq 20\%$ of average monthly personnel rate during the 1st reporting period:

WP	Justification for deviation
ENV	Higher average monthly personnel costs than planned because of more senior experts working on the project during the 1 st reporting period. This will be balanced in future reporting periods.
ULA	Higher average monthly personnel costs than planned due to a number of factors: In 2024, the salaries of professors at the University of L'Aquila were adjusted in line with ISTAT tables, leading to a general increase in salaries. There have also been career advancements for two researchers, who became associate professors. Other researchers received salary increases. Additionally, the two-year contract for young researchers, which has a lower monthly average than that of teachers and would balance the salary rate, has not yet been activated.
JRG	Higher average monthly personnel costs due to an error in the calculation during the proposal phase. All necessary work in the project will be carried out with the budget available.
KhNU	Lower average monthly personnel costs result mainly from the depreciation of the Ukrainian hryvnia, as all salaries are paid in UAH and their EUR-equivalent value is significantly reduced when converted using the official monthly exchange rate. The involvement of KhNU in the project increases substantially after Month 13 with the start of key tasks and deliverables, therefore, higher personnel costs are expected in the next reporting periods.
LYKUML	The calculation of the actual personnel costs led to a lower monthly average than originally estimated, which is also due to the falling exchange rate.
ME IUD	The calculation of the actual personnel costs led to a lower monthly average than originally estimated, which is also due to the falling exchange rate.

5.2.1. Unforeseen subcontracting

Not applicable.

5.2.2. Unforeseen use of in kind contributions

Not applicable.

#§WRK-PLA-WP§#